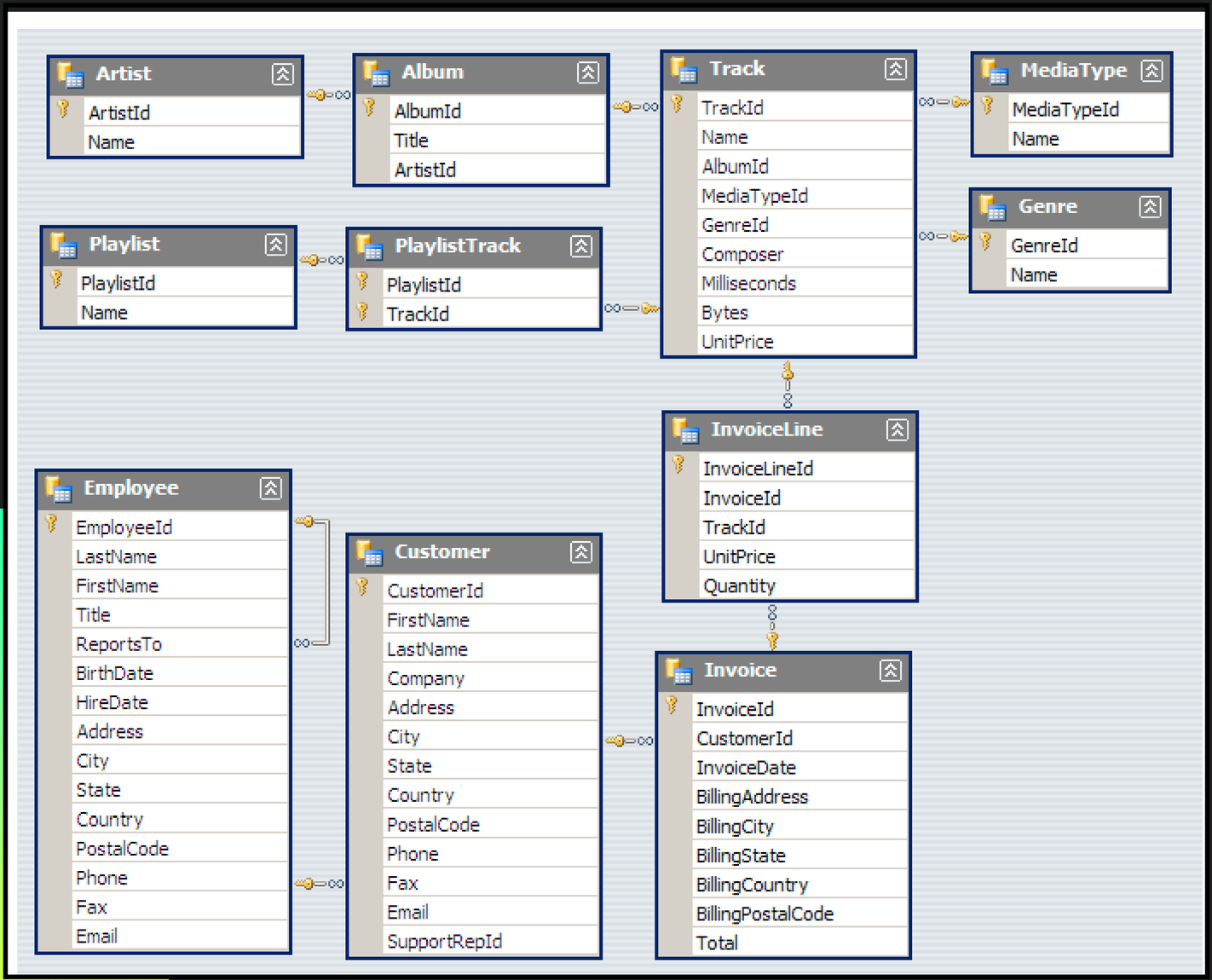
DIGITAL MUSIC STORE DATA ANALYSIS USING SQL

OBJECTIVE

We will analyze the dataset music store database with SQL and help the store to understand the business growth by answering Question’s

SQL Server: Microsoft sql database Managemnet studio 19

DATABASE SCHEMA



A database schema is essentially a blueprint for how a database is structured. It defines the organization of data and how different pieces of data relate to each other. In simple terms, it's a plan that shows how data is stored and organized within a database.

1. Who is the senior most employee based on job title?

```sql
SELECT
    TOP 1
    employee_id,
    first_name,
    last_name,
    title,
    levels,
    address,
    city
FROM
    employee
ORDER BY
    levels DESC
```

Results | Messages

	employee_id	first_name	last_name	title	levels	address	city
1	9	Mohan	Madan	Senior General Manager	L7	1008 Vrinda Ave MT	Edmonton

2. Which countries have the most Invoices?

```sql
SELECT
    billing_country,
    COUNT(*) AS No_of_Invoices
FROM
    invoice
GROUP BY
    billing_country
ORDER BY
    No_of_Invoices DESC;
```

Results | Messages

	billing_country	No_of_Invoices
1	USA	131
2	Canada	76
3	Brazil	61
4	France	50
5	Germany	41
6	Czech Republic	30
7	Portugal	29
8	United Kingdom	28

3. What are top 3 values of total invoice?

```sql
SELECT
    TOP 3
    total,
    invoice_id,
    customer_id

FROM
    invoice
ORDER BY
    total DESC
```

Results | Messages

	total	invoice_id	customer_id
1	23.7600002288818	183	42
2	19.7999992370605	31	3
3	19.7999992370605	92	32

4. Which city has the best customers? We would like to throw a promotional Music Festival in the city we made the most money. Write a query that returns one city that has the highest sum of invoice totals. Return both the city name & sum of all invoice totals

```sql
SELECT
    billing_city,
    SUM(total) AS 'Invoice Total'
FROM
    invoice
GROUP BY
    billing_city
ORDER BY
    'Invoice Total' DESC;
```

Results | Messages

	billing_city	Invoice Total
1	Prague	273.240000247955
2	Mountain View	169.290001153946
3	London	166.319998502731
4	Berlin	158.399999380112
5	Paris	151.470000505447
6	São Paulo	129.689999818802
7	Dublin	114.839999437332
8	Delhi	111.869999647141

5. Who is the best customer? The customer who has spent the most money will be declared the best customer. Write a query that returns the person who has spent the most money

```sql
SELECT
    TOP 1
    customer.first_name,
    customer.last_name,
    customer.customer_id,
    SUM(invoice.total) AS Total_invoice
FROM
    customer
JOIN
    invoice ON customer.customer_id = invoice.customer_id
GROUP BY
    customer.customer_id,
    customer.first_name,
    customer.last_name
ORDER BY
    Total_invoice DESC
```

Results | Messages

	first_name	last_name	customer_id	Total_invoice
1	František	Wichterlová	5	144.539998054504

6. Write query to return the email, first name, last name, & Genre of all Rock Music listeners. Return your list ordered alphabetically by email starting with A

```sql
SELECT DISTINCT
    customer.email,
    customer.first_name,
    customer.last_name
FROM
    customer
JOIN
    invoice ON customer.customer_id = invoice.customer_id
JOIN
    invoice_line ON invoice.invoice_id = invoice_line.invoice_id
JOIN
    track ON invoice_line.track_id = track.track_id
JOIN
    genre ON track.genre_id = genre.genre_id
WHERE
    genre.name LIKE 'ROCK'
ORDER BY
    customer.email;
```

Results | Messages

	email	first_name	last_name
1	aaronmitchell@yahoo.ca	Aaron	Mitchell
2	alero@uol.com.br	Alexandre	Rocha
3	astrid.gruber@apple.at	Astrid	Gruber
4	bjorn.hansen@yahoo.no	Bjørn	Hansen
5	camille.bernard@yaho...	Camille	Bernard
6	daan_peeters@apple.be	Daan	Peeters
7	diego.gutierrez@yahoo...	Diego	Gutiérrez
8	dmiller@comcast.com	Dan	Miller
9	dominiquelefebvre@g...	Dominique	Lefebvre

7. Let's invite the artists who have written the most rock music in our dataset. Write a query that returns the Artist name and total track count of the top 10 rock bands

```sql
SELECT
    TOP 10
    a.artist_id,
    a.name,
    COUNT(t.track_id) AS Total_Track_Count
FROM
    track t
JOIN
    album al ON t.album_id = al.album_id
JOIN
    artist a ON a.artist_id = al.artist_id
JOIN
    genre g ON g.genre_id = t.genre_id
WHERE
    g.name = 'ROCK'
GROUP BY
    a.artist_id,
```

Results | Messages

	artist_id	name	Total_Track_Count
1	22	Led Zeppelin	114
2	150	U2	112
3	58	Deep Purple	92
4	90	Iron Maiden	81
5	118	Pearl Jam	54
6	152	Van Halen	52
7	51	Queen	45
8	142	The Rolling Stones	41
9	76	Creedence Clearwater Revival	40
10	52	Kiss	35

8. Return all the track names that have a song length longer than the average song length. Return the Name and Milliseconds for each track. Order by the song length with the longest songs listed first

```sql
SELECT
    name,
    milliseconds
FROM
    track
WHERE
    milliseconds > (SELECT AVG(milliseconds) FROM track)
ORDER BY
    milliseconds DESC;
```

Results | Messages

	name	milliseconds
1	Occupation / Precipice	5286953
2	Through a Looking Glass	5088838
3	Greetings from Earth, Pt. 1	2960293
4	The Man With Nine Lives	2956998
5	Battlestar Galactica, Pt. 2	2956081
6	Battlestar Galactica, Pt. 1	2952702
7	Murder On the Rising Star	2935894
8	Battlestar Galactica, Pt. 3	2927802
9	Take the Celestra	2927677
10	Fire In Space	2926593
11	The Long Patrol	2925008
12	The Magnificent Warriors	2924716

9. Find how much amount spent by each customer on artists? Write a query to return customer name, artist name and total spent

```sql
WITH best_selling_artist AS (
        SELECT TOP 1
        artist.artist_id as 'artist id',
        artist.name as 'artist name',
        SUM(invoice_line.unit_price*invoice_line.quantity) AS 'total_sales'
        FROM invoice_line
        JOIN track ON track.track_id = invoice_line.track_id
        JOIN album ON album.album_id = track.album_id
        JOIN artist ON artist.artist_id = album.artist_id
        GROUP BY artist.artist_id, artist.name
        ORDER BY 3 DESC )
SELECT
    customer.customer_id,
    customer.first_name,
    customer.last_name,
    SUM(invoice_line.unit_price*invoice_line.quantity) AS 'amount_spent',
    best_selling_artist.[artist name]
    FROM invoice
    JOIN customer ON customer.customer_id =invoice.customer_id
    JOIN invoice_line ON invoice_line.invoice_id = invoice.invoice_id
    JOIN track ON track.track_id = invoice_line.track_id
    JOIN album ON album.album_id = track.album_id
    JOIN best_selling_artist ON best_selling_artist.[artist id] = album.artist_id
    GROUP BY customer.customer_id,customer.first_name,customer.last_name,best_selling_artist.[artist name]
    ORDER BY [amount_spent] DESC
```

Results | Messages

	customer_id	first_name	last_name	amount_spent	artist name
1	46	Hugh	O'Reilly	27.7200002670288	Queen
2	38	Niklas	Schröder	18.8100001811981	Queen
3	3	François	Tremblay	17.8200001716614	Queen
4	34	João	Fernand...	16.8300001621246	Queen
5	41	Marc	Dubois	11.8800001144409	Queen
6	53	Phil	Hughes	11.8800001144409	Queen
7	47	Lucas	Mancini	10.8900001049042	Queen
8	33	Ellie	Sullivan	10.8900001049042	Queen
9	5	František	Wichterl...	3.96000003814697	Queen
10	20	Dan	Miller	3.96000003814697	Queen
11	31	Martha	Silk	2.97000002861023	Queen
12	23	John	Gordon	2.97000002861023	Queen

10. We want to find out the most popular music Genre for each country. We determine the most popular genre as the genre with the highest amount of purchases. Write a query that returns each country along with the top Genre. For countries where the maximum number of purchases is shared return all Genres

```sql
WITH popular_genre AS (
    SELECT COUNT(invoice_line.quantity) AS purchases,
        customer.country,
        genre.name AS genre_name,
        genre.genre_id,
        ROW_NUMBER() OVER
        (PARTITION BY customer.country ORDER BY COUNT(invoice_line.quantity) DESC)
        AS row_num
    FROM
    invoice_line
    JOIN invoice ON invoice.invoice_id = invoice_line.invoice_id
    JOIN customer ON customer.customer_id = invoice.customer_id
    JOIN track ON track.track_id = invoice_line.track_id
    JOIN genre ON genre.genre_id = track.genre_id
    GROUP BY
        customer.country, genre.name, genre.genre_id
)
SELECT purchases,country,genre_name
FROM
    popular_genre
WHERE
    row_num = 1;
```

Results | Messages

	purchases	country	genre_name
1	17	Argentina	Alternative & Punk
2	34	Australia	Rock
3	40	Austria	Rock
4	26	Belgium	Rock
5	205	Brazil	Rock
6	333	Canada	Rock
7	61	Chile	Rock
8	143	Czech Republic	Rock
9	24	Denmark	Rock
10	46	Finland	Rock
11	211	France	Rock
12	194	Germany	Rock
13	44	Hungary	Rock
14	102	India	Rock

11. Write a query that determines the customer that has spent the most on music for each country. Write a query that returns the country along with the top customer and how much they spent. For countries where the top amount spent is shared, provide all customers who spent this amount

```sql
WITH customer_with_country AS (
    SELECT
        customer.customer_id,
        customer.first_name,
        customer.last_name,
        invoice.billing_country,
        SUM(invoice.total) AS total_spending,
        ROW_NUMBER() OVER (PARTITION BY invoice.billing_country
        ORDER BY SUM(invoice.total) DESC) AS rowNo
    FROM
        invoice
    JOIN
        customer ON customer.customer_id = invoice.customer_id
    GROUP BY
        customer.customer_id, customer.first_name, customer.last_name, invoice.billing_country
    ORDER BY
        billing_country asc , total_spending desc offset 0 rows
)
SELECT customer_id,first_name,last_name,billing_country,total_spending
FROM
    customer_with_country
WHERE
    rowNo = 1;
```

Results | Messages

	customer_id	first_name	last_name	billing_country	total_spending
1	56	Diego	Gutiérrez	Argentina	39.5999991893768
2	55	Mark	Taylor	Australia	81.1800003051758
3	7	Astrid	Gruber	Austria	69.299998998642
4	8	Daan	Peeters	Belgium	60.3899998664856
5	1	Luís	Gonçalves	Brazil	108.899999380112
6	3	François	Tremblay	Canada	99.9899981021881
7	57	Luis	Rojas	Chile	97.0199997425079
8	5	František	Wichterlová	Czech Republic	144.539998054504
9	9	Kara	Nielsen	Denmark	37.6199996471405

12. Retrieve the most expensive track for each genre

```
SELECT g.Name AS GenreName, t.Name AS TrackName, MAX(t.unit_price) AS MaxPrice
FROM track t
JOIN genre g ON t.Genre_Id = g.Genre_Id
GROUP BY g.Name, t.Name;
```

Results | Messages

	GenreName	TrackName	MaxPrice
1	TV Shows	"?"	1.99000000953674
2	Rock	"40"	0.990000009536743
3	Classical	"Eine Kleine Nachtmusik" Serenade In G, K. 525:...	0.990000009536743
4	Alternative & Punk	#1 Zero	0.990000009536743
5	Pop	#9 Dream	0.990000009536743
6	Jazz	'Round Midnight	0.990000009536743
7	Metal	(Anesthesia) Pulling Teeth	0.990000009536743

13. List all playlists and the number of tracks in each playlist

```
SELECT pl.Name AS PlaylistName, COUNT(plt.Track_Id) AS NumberOfTracks
FROM playlist pl
LEFT JOIN playlist_track plt ON pl.Playlist_Id = plt.Playlist_Id
GROUP BY pl.Name order by NumberOfTracks DESC
```

Results | Messages

	PlaylistName	NumberOfTracks
1	Music	6580
2	90's Music	1477
3	TV Shows	426
4	Classical	75
5	Brazilian Music	39
6	Heavy Metal Classic	26
7	Classical 101 - Deep Cuts	25

14. Find the average price of tracks in each album

```sql
SELECT a.Title AS AlbumTitle, AVG(t.Unit_Price) AS AveragePrice
FROM album a
JOIN track t ON a.album_id = t.Album_Id
GROUP BY a.Title
```

Results | Messages

	AlbumTitle	AveragePrice
1	...And Justice For All	0.990000009536743
2	[1997] Black Light Syndrome	0.990000009536743
3	20th Century Masters - The Millennium Collectio...	0.990000009536743
4	A Copland Celebration, Vol. I	0.990000009536743
5	A Matter of Life and Death	0.990000009536743
6	A Real Dead One	0.990000009536743
7	A Real Live One	0.990000009536743

15. Find the top 3 most recent invoices and their total sales

```sql
SELECT TOP 3 i.invoice_id, i.Invoice_Date, SUM(il.Unit_Price * il.Quantity) AS TotalSales
FROM invoice i
JOIN invoice_line il ON i.Invoice_Id = il.Invoice_Id
GROUP BY i.Invoice_Id, i.Invoice_Date
ORDER BY i.Invoice_Date DESC
```

Results | Messages

	invoice_id	Invoice_Date	TotalSales
1	614	2020-12-30 00:00:00.0000000	12.8700001239777
2	613	2020-12-29 00:00:00.0000000	8.91000008583069
3	611	2020-12-27 00:00:00.0000000	1.98000001907349

16. Get the genre with the highest total sales

```sql
SELECT TOP 1 g.Name AS GenreName
FROM genre g
JOIN track t ON g.Genre_Id = t.Genre_Id
JOIN invoice_line il ON t.Track_Id = il.Track_Id
GROUP BY g.Name
ORDER BY SUM(il.Unit_Price * il.Quantity) DESC
```

Results | Messages

	GenreName
1	Rock

17 Retrieve the list of tracks and their composers with no sales

```sql
SELECT t.Name AS TrackName, t.Composer
FROM track t
LEFT JOIN invoice_line il ON t.Track_Id = il.Track_Id
WHERE il.Track_Id IS NULL
```

Results | Messages

	TrackName	Composer
1	Your Time Has Come	Cornell, Commerford, Morello, Wilk
2	Be Yourself	Cornell, Commerford, Morello, Wilk
3	Heaven's Dead	Cornell, Commerford, Morello, Wilk
4	Man Or Animal	Cornell, Commerford, Morello, Wilk
5	Yesterday To Tomorrow	Cornell, Commerford, Morello, Wilk
6	Money	Berry Gordy, Jr./Janie Bradford
7	Long Tall Sally	Enotris Johnson/Little Richard/Robert "Bumps" Bl...

18 .Find all customers who have purchased tracks from a specific genre

```sql
SELECT DISTINCT c.First_Name, c.Last_Name
FROM customer c
JOIN invoice i ON c.Customer_Id = i.Customer_Id
JOIN invoice_line il ON i.Invoice_Id = il.Invoice_Id
JOIN track t ON il.Track_Id = t.Track_Id
WHERE t.Genre_Id = (SELECT Genre_Id FROM genre WHERE Name = 'Jazz')
```

Results | Messages

	First_Name	Last_Name
1	Aaron	Mitchell
2	Astrid	Gruber
3	Bjørn	Hansen
4	Camille	Bernard
5	Daan	Peeters
6	Dan	Miller
7	Eduardo	Martins

19. Show the total number of tracks and their total sales in each genre

```sql
SELECT g.Name AS GenreName, COUNT(t.Track_Id) AS TotalTracks, SUM(il.Unit_Price * il.quantity) AS TotalSales
FROM genre g
JOIN track t ON g.Genre_Id = t.Genre_Id
LEFT JOIN invoice_line il ON t.Track_Id = il.Track_Id
GROUP BY g.Name;
```

Results | Messages

	GenreName	TotalTracks	TotalSales
1	Pop	86	62.3700006008148
2	Hip Hop/Rap	47	32.6700003147125
3	Bossa Nova	15	NULL
4	World	28	NULL
5	Comedy	17	NULL
6	Rock	3017	2608.65002512932
7	Drama	64	0.99000009536743

BUSINESS FINDINGS

- **Top Customers:** Identify high-value customers to target for loyalty programs and personalized offers.
- **Customer Retention:** Track repeat purchases to assess and improve customer loyalty.
- **Popular Music Tracks:** Highlight best-selling tracks to boost inventory management and promotions.
- **Sales Trends:** Analyze sales patterns over time to identify seasonal trends and peak periods.
- **Revenue by Genre:** Focus on profitable genres to tailor marketing strategies and manage inventory.
- **Top Artists:** Promote high-revenue artists to enhance visibility and sales.
- **Artist Popularity Trends:** Adjust marketing and inventory based on artist popularity shifts.

- **Stock Levels:** Monitor inventory to prevent stockouts and overstock situations.

- **Order Fulfillment Efficiency:** Improve fulfillment processes to enhance customer satisfaction.

- **Promotion Effectiveness:** Evaluate promotional impact to refine future marketing campaigns.

- **Customer Demographics:** Tailor marketing efforts based on customer demographics for better targeting.

- **Overall Revenue Analysis:** Assess total revenue to gauge business performance and growth.

- **Customer Acquisition Costs:** Evaluate acquisition costs against customer lifetime value for marketing efficiency.

SUMMARY

By leveraging these insights, a digital music store can make informed decisions regarding marketing strategies, inventory management, customer engagement, and overall business strategy. These insights support optimizing operations, enhancing customer satisfaction, and driving revenue growth.

THANK YOU